
KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

arxiv.org/abs/2504.03553

ML Research | LLM Agents

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Efficiency

Current methods inject gold trajectories, external feedback, or domain knowledge regardless of situational need — leading to fragile planning and high inference costs.

Trajectory Imitation

ReAct

- Directly mimics gold trajectories
- No situational awareness
- Pattern collapse on novel inputs

Trial-and-Error

Reflexion

- Iterative feedback loops
- External critique regardless of need
- High computational overhead

Knowledge-Augmented

KnowAgent, WKM, ExpeL

- Injects domain knowledge at every step
- 100% knowledge usage — wasteful
- Can hurt rather than help performance

Humans possess situational self-awareness. KnowSelf brings this metacognitive ability to LLM agents.

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking

At each decision step, KnowSelf classifies the agent's cognitive situation using a heuristic criterion.

Fast Thinking

[FAST]

- Agent directly produces the correct action
- Minimal deliberation required
- Predicted action matches gold action

Slow Thinking

[REFLECT]

- Initially produces an incorrect action
- Can self-correct through reflection
- Reflection fixes the prediction

Knowledgeable Thinking

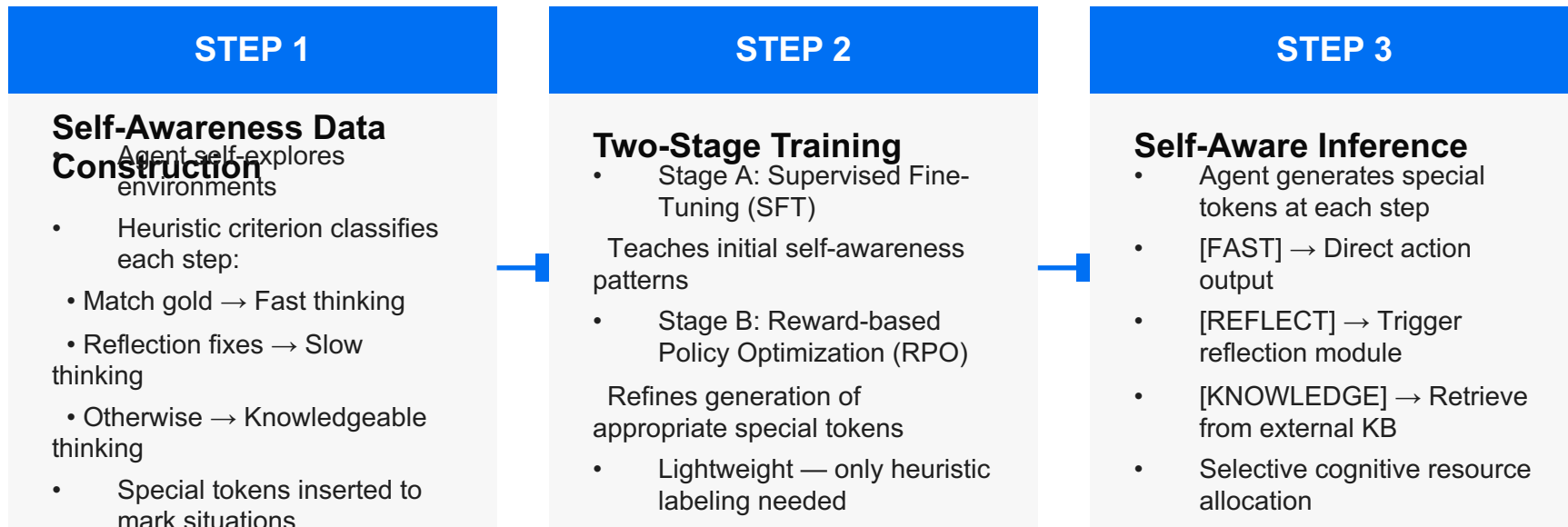
[KNOWLEDGE]

- Cannot produce correct action even after reflection
- Must consult external knowledge base
- Knowledge retrieval is necessary

Decision Flow:

Predict action → Match gold? → YES: [FAST] | NO: Reflect → Fixed? → YES: [REFLECT] | NO: [KNOWLEDGE]

KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference



Framework diagram: figure_3.jpg

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task: Put a hot mug in cabinet

OpenAI-O1 (Baseline)

Incorrectly tries to take an egg from the microwave after opening it — fails to recognize the mug needs heating first.

KnowSelf (Llama-8B)

Uses [REFLECT] token to realize it should heat the mug using the open microwave — corrects its own plan autonomously.

[REFLECT]

Task: Put a saltshaker in drawer

DeepSeek-R1 (Baseline)

Reflects that the drawer is open and tries to close it first — action fails, getting stuck on irrelevant state changes.

KnowSelf (Llama-8B)

Applies [KNOWLEDGE] token noting the agent should place the object without removing unrelated items — succeeds directly.

[KNOWLEDGE]

Qualitative comparison: figure_1.jpg

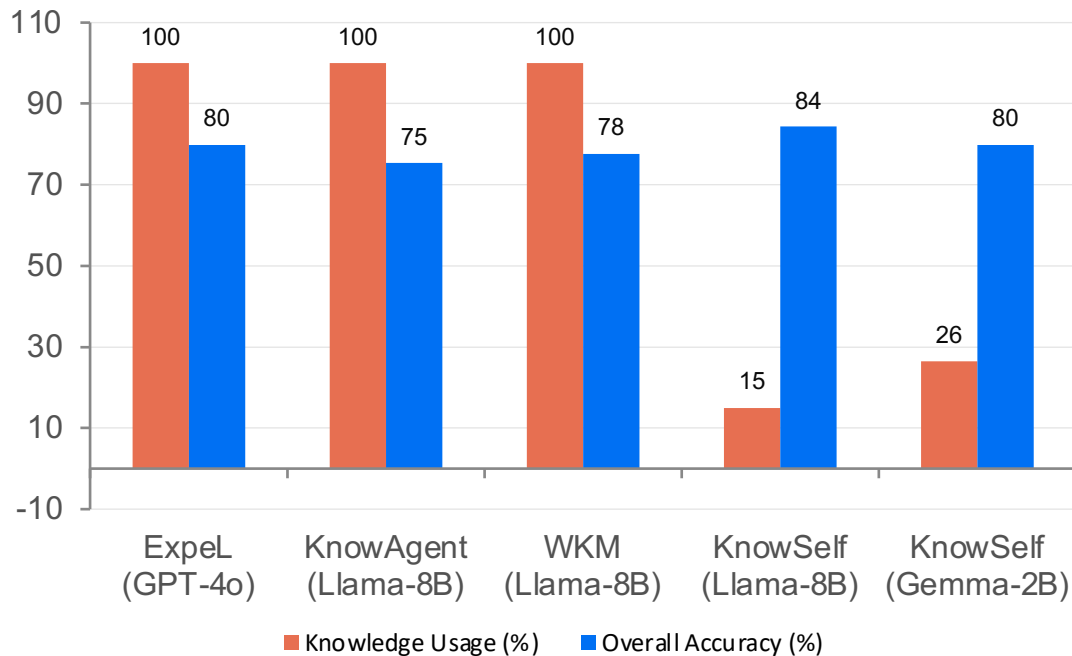
KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Success rates (%) across six ALFWorld task types. Know% = proportion of steps using external knowledge.

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

KnowSelf (Llama-8B) surpasses GPT-4o ExpeL (84.33% vs 79.85%) while using only 15% knowledge

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge Methods



15.01%

Knowledge used by KnowSelf (Llama-8B)

84.33%

Overall accuracy — best across all methods

5.6×

Less knowledge than 100% methods, yet higher accuracy

Even Gemma-2B (79.85%) matches GPT-4o Expel — selective knowledge scales across model sizes.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

- 01 Indiscriminate knowledge injection is wasteful and suboptimal — blanket application hurts agent performance.
- 02 Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.
- 03 KnowSelf with Llama-8B outperforms strong baselines including GPT-4o ReAct (84.33% vs 76.12%) and GPT-4o ExpeL (79.85%).
- 04 Only 15–26% knowledge usage needed versus 100% for competing methods — a 5.6× efficiency gain.
- 05 The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%,

Future Directions: Extend to multi-agent systems, dynamic knowledge bases, and real-world robotics tasks.